AWS VPC Peering Implementation & Verification Report

Cross-VPC Interconnectivity, Security Group Isolation, and Failure Recovery Analysis

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Batch No. : 1

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1. Executive Summary & Project Overview

This report details the architectural design, deployment, verification, and teardown of an AWS VPC Peering infrastructure connecting two isolated Virtual Private Clouds (VPC A and VPC B) in the **ap-southeast-2 (Sydney)** region. The objective was to establish secure, low-latency private connectivity between resources in separate network boundaries while maintaining strict routing controls and security isolation.

2. Architecture Diagram & Resource Inventory

Architecture Overview:

1. **VPC A (10.10.0.0/16):** Contains one public subnet (ewb-a-public - 10.10.1.0/24) housing the Bastion host (ewb-bastion), and one private subnet (ewb-a-private - 10.10.2.0/24) housing Application Instance A (ewb-app-a).
2. **VPC B (10.20.0.0/16):** Contains one private subnet (ewb-b-private - 10.20.1.0/24) housing Application Instance B (ewb-app-b).
3. **VPC Peering Connection (pcx-046f17b39297c1c8d):** Enables direct private IPv4 routing between VPC A and VPC B CIDRs.

EC2 Resource Inventory:

Instance Name	Instance ID	Subnet	Private IP	VPC ID
ewb-bastion	i-0ff13baec33357ffc	ewb-a-public	10.10.1.10	vpc-0a3ae026fbb3e376f
ewb-app-a	i-07150b1244b947b39	ewb-a-private	10.10.2.10	vpc-0a3ae026fbb3e376f
ewb-app-b	i-011fa5da79510040d	ewb-b-private	10.20.1.10	vpc-0681b961980067d63

3. Subnets, Route Tables & Security Groups Configuration

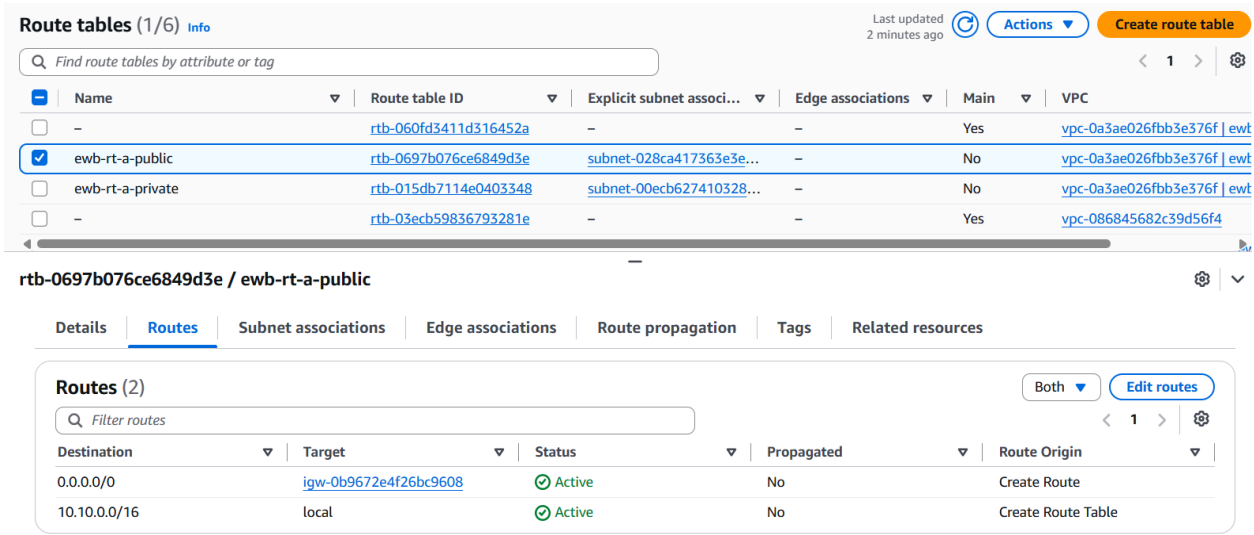
Subnet Configuration:

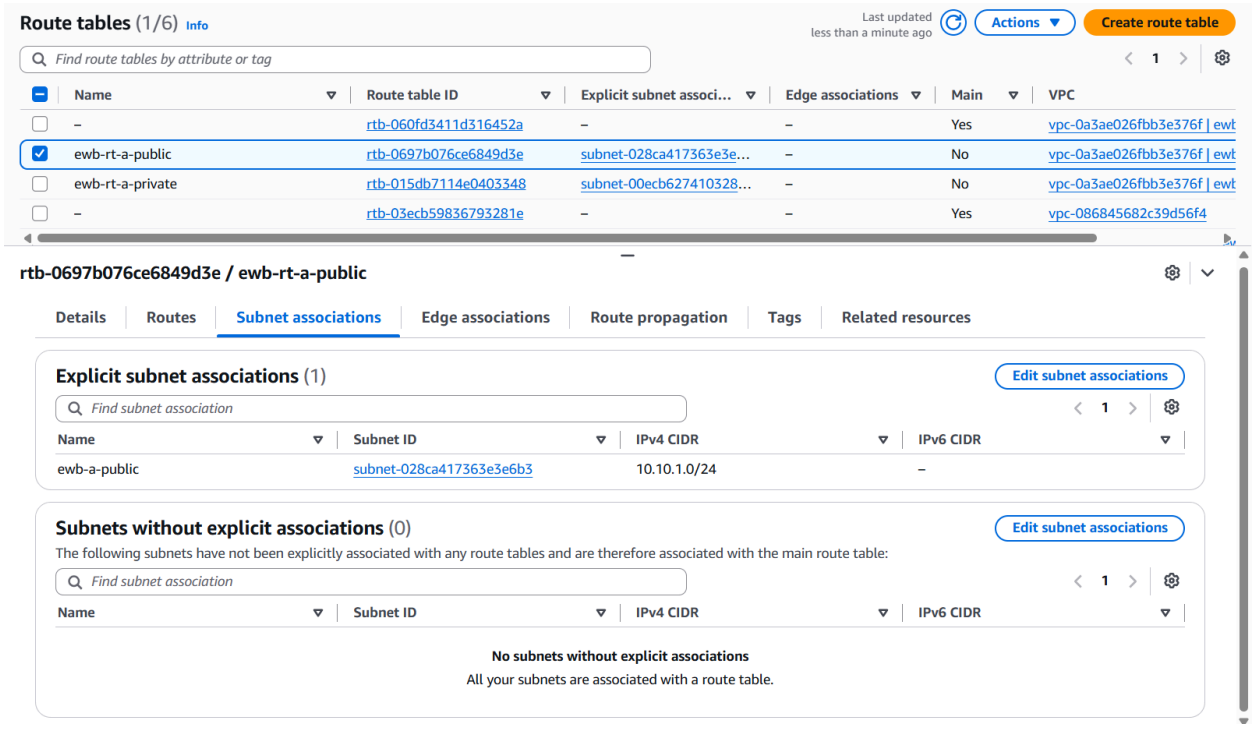
4. **ewb-a-public (subnet-028ca417363e3e6b3):** 10.10.1.0/24 in VPC A (ap-southeast-2a)
5. **ewb-a-private (subnet-00ecb627410328c3c):** 10.10.2.0/24 in VPC A (ap-southeast-2a)
6. **ewb-b-private (subnet-06246468f1c2ad4d8):** 10.20.1.0/24 in VPC B (ap-southeast-2a)

Route Table Specifications:

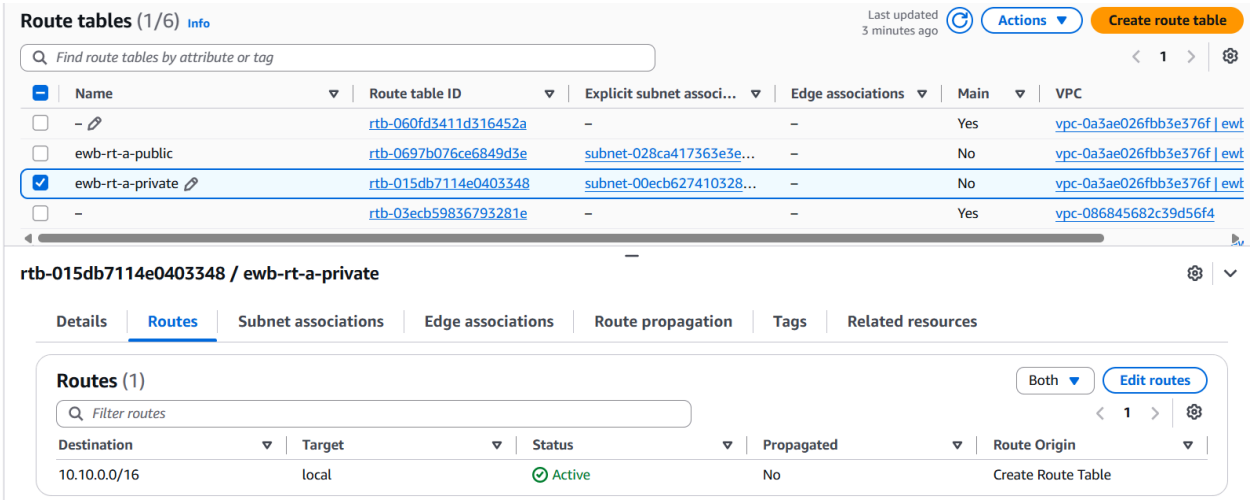
7. **ewb-rt-a-public (rtb-0697b076ce6849d3e):** Routes 0.0.0.0/0 to IGW (igw-0b9672e4f26bc9608), 10.10.0.0/16 local, 10.20.0.0/16 to Peering Connection (pcx-046f17b39297c1c8d).
8. **ewb-rt-a-private (rtb-015db7114e0403348):** Routes 10.10.0.0/16 local, 10.20.0.0/16 to Peering Connection (pcx-046f17b39297c1c8d).
9. **ewb-rt-b-private (rtb-0bccfcf9b8c6cf4d8):** Routes 10.20.0.0/16 local, 10.10.0.0/16 to Peering Connection (pcx-046f17b39297c1c8d).

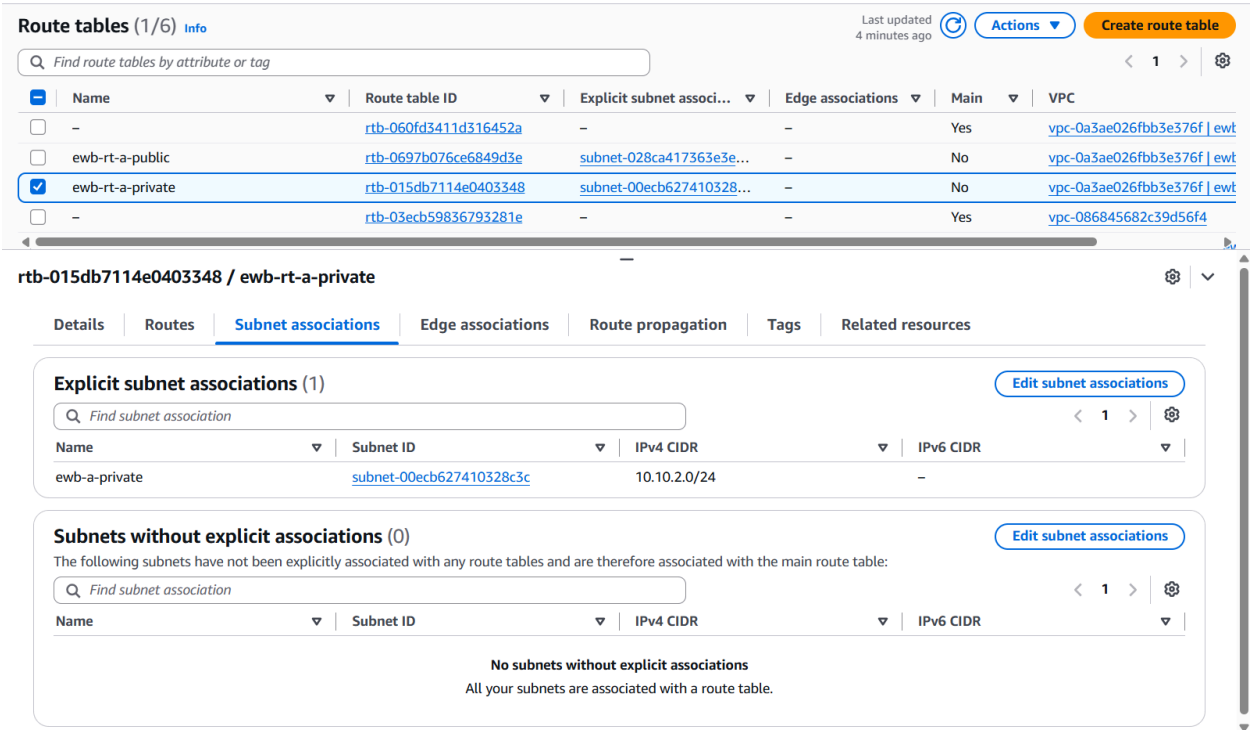
Subnet Associations & Pre-Peering Setup Verification:



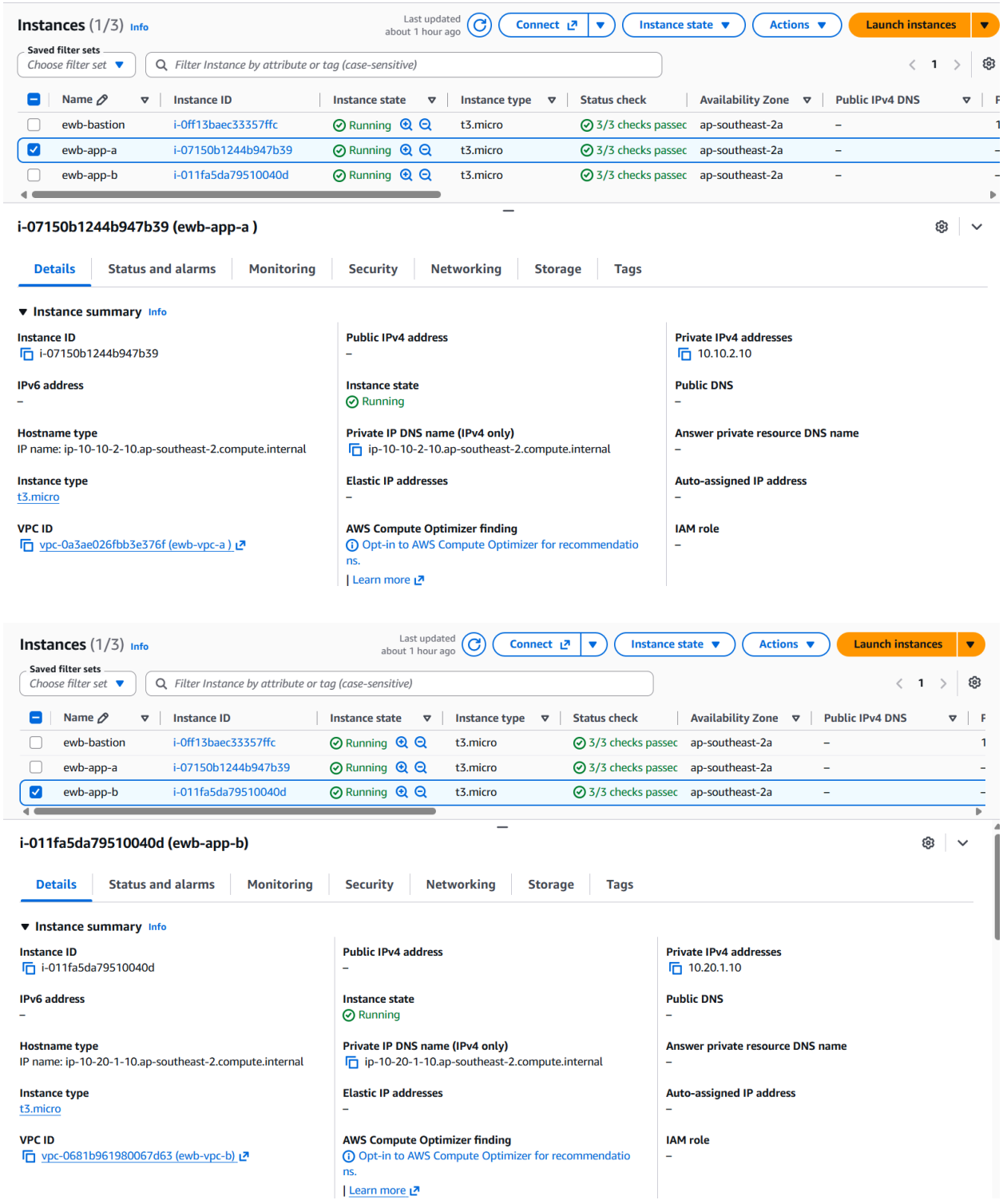


Private Subnet Route Tables (Pre-Peering):





EC2 Instance Status & IP Allocations:



Instances (1/3) Info

Last updated about 1 hour ago | Connect | Instance state | Actions | Launch instances

Saved filter sets: Choose filter set

Filter Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
ewb-bastion	i-0ff13baec33357ffc	Running	t3.micro	3/3 checks passec	ap-southeast-2a	–
ewb-app-a	i-07150b1244b947b39	Running	t3.micro	3/3 checks passec	ap-southeast-2a	–
ewb-app-b	i-011fa5da79510040d	Running	t3.micro	3/3 checks passec	ap-southeast-2a	–

i-07150b1244b947b39 (ewb-app-a)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

Instance summary Info

Instance ID
i-07150b1244b947b39

IPv6 address
–

Hostname type
IP name: ip-10-10-2-10.ap-southeast-2.compute.internal

Instance type
t3.micro

VPC ID
vpc-0a3ae026fbb3e376f (ewb-vpc-a)

Public IPv4 address
–

Instance state
Running

Private IP DNS name (IPv4 only)
ip-10-10-2-10.ap-southeast-2.compute.internal

Elastic IP addresses
–

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Private IPv4 addresses
10.10.2.10

Public DNS
–

Answer private resource DNS name
–

Auto-assigned IP address
–

IAM role
–

Instances (1/3) Info

Last updated about 1 hour ago | Connect | Instance state | Actions | Launch instances

Saved filter sets: Choose filter set

Filter Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
ewb-bastion	i-0ff13baec33357ffc	Running	t3.micro	3/3 checks passec	ap-southeast-2a	–
ewb-app-a	i-07150b1244b947b39	Running	t3.micro	3/3 checks passec	ap-southeast-2a	–
ewb-app-b	i-011fa5da79510040d	Running	t3.micro	3/3 checks passec	ap-southeast-2a	–

i-011fa5da79510040d (ewb-app-b)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

Instance summary Info

Instance ID
i-011fa5da79510040d

IPv6 address
–

Hostname type
IP name: ip-10-20-1-10.ap-southeast-2.compute.internal

Instance type
t3.micro

VPC ID
vpc-0681b961980067d63 (ewb-vpc-b)

Public IPv4 address
–

Instance state
Running

Private IP DNS name (IPv4 only)
ip-10-20-1-10.ap-southeast-2.compute.internal

Elastic IP addresses
–

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Private IPv4 addresses
10.20.1.10

Public DNS
–

Answer private resource DNS name
–

Auto-assigned IP address
–

IAM role
–

Security Group Inbound Rules (Documented Pre-Cleanup):

- ewb-sg-bastion (VPC A): Type: SSH (TCP 22) | Source: 157.49.42.224/32
- ewb-sg-app-a (VPC A): Type: SSH (TCP 22) | Source: 10.10.1.10/32
- ewb-sg-app-a (VPC A): Type: Custom ICMP - IPv4 (Echo Request) | Source: 10.20.1.10/32
- ewb-sg-app-b (VPC B): Type: SSH (TCP 22) | Source: 10.10.1.10/32
- ewb-sg-app-b (VPC B): Type: Custom ICMP - IPv4 (Echo Request) | Source: 10.10.2.10/32

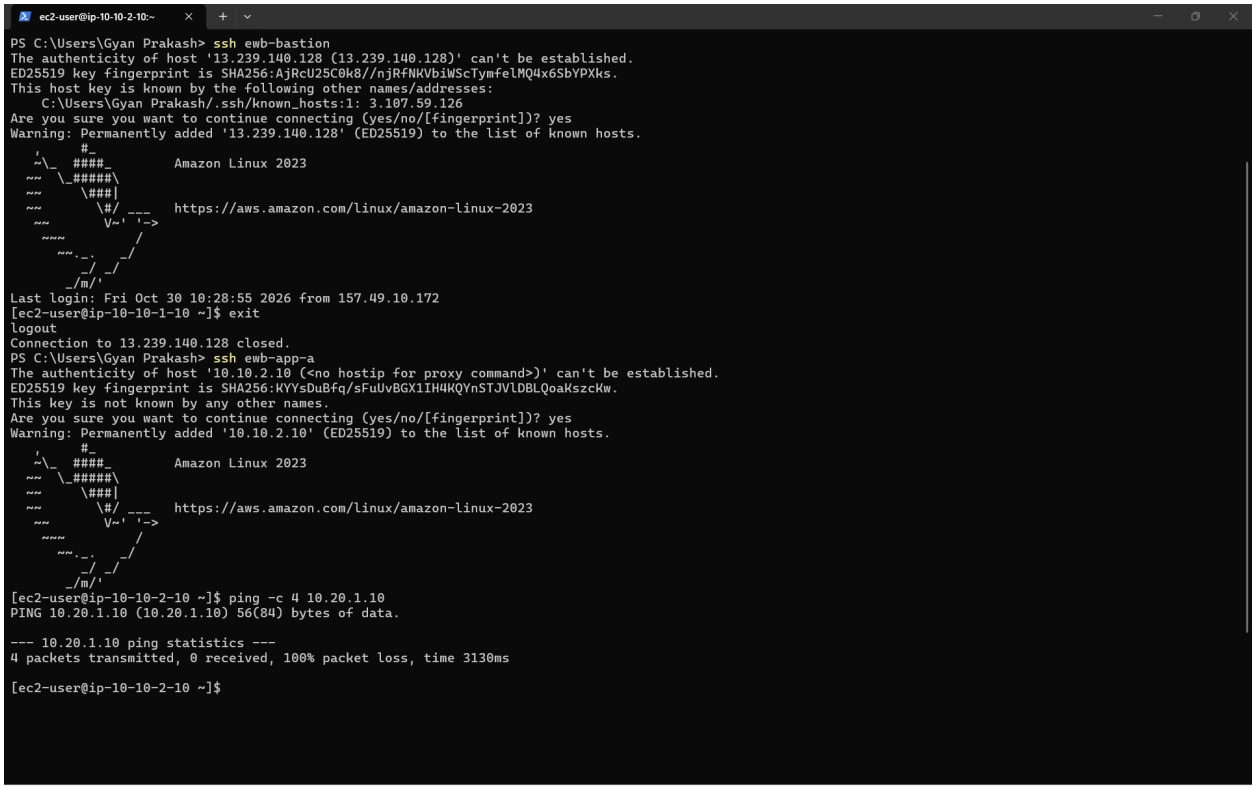
Note: Security Groups were successfully configured to allow strict access as proven by the Validation Tests in Section 4. The resources were deleted during the final cleanup phase.

4. Validation Tests & Execution Evidence

A. Baseline Failure Verification (Pre-Peering)

Before establishing active peering routes, connectivity between ewb-app-a (10.10.2.10) and ewb-app-b (10.20.1.10) was tested using ICMP ping. As expected, packets were dropped (100% packet loss) due to missing route table entries for cross-VPC traffic.

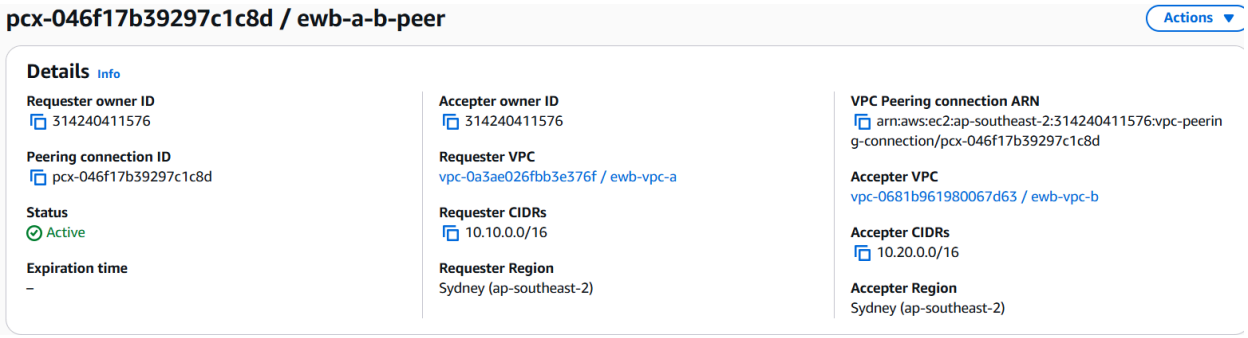
Baseline Test Evidence (Test_01_Baseline_Failure.png):



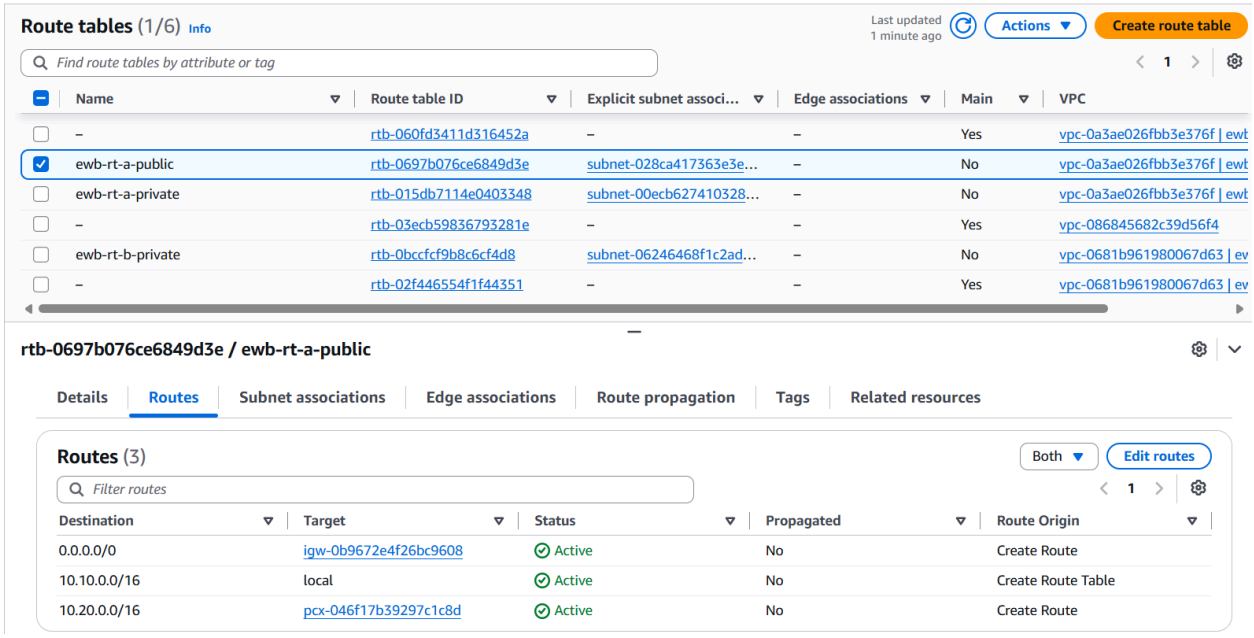
B. Active Peering Connection Details

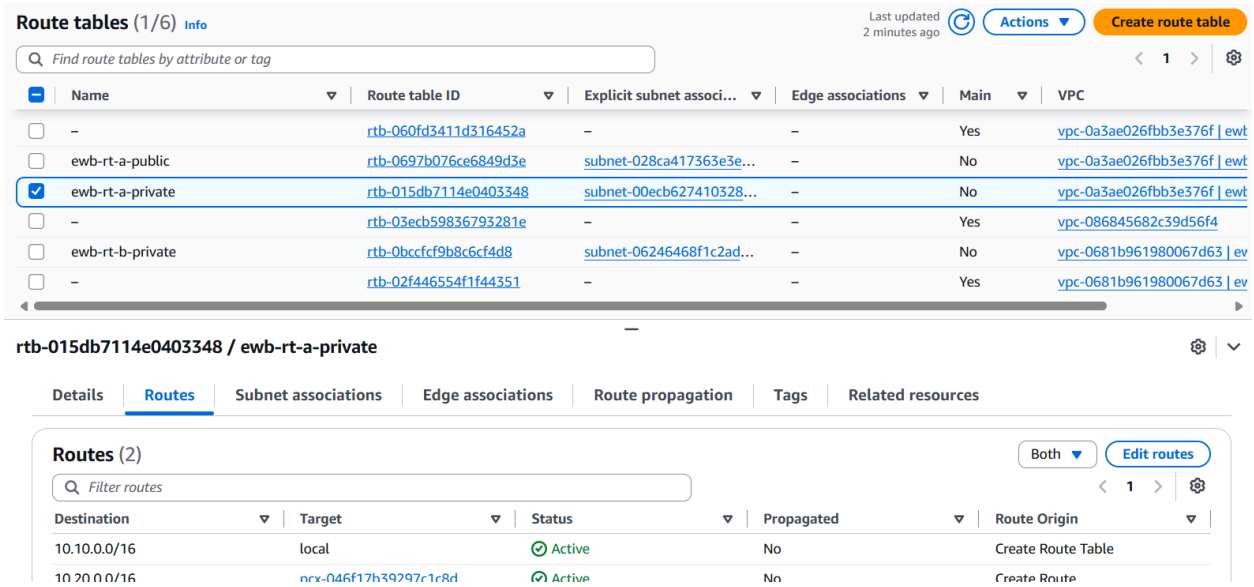
VPC Peering connection pcx-046f17b39297c1c8d (ewb-a-b-peer) was created between Requester VPC A (10.10.0.0/16) and Accepter VPC B (10.20.0.0/16) in region ap-southeast-2 and accepted.

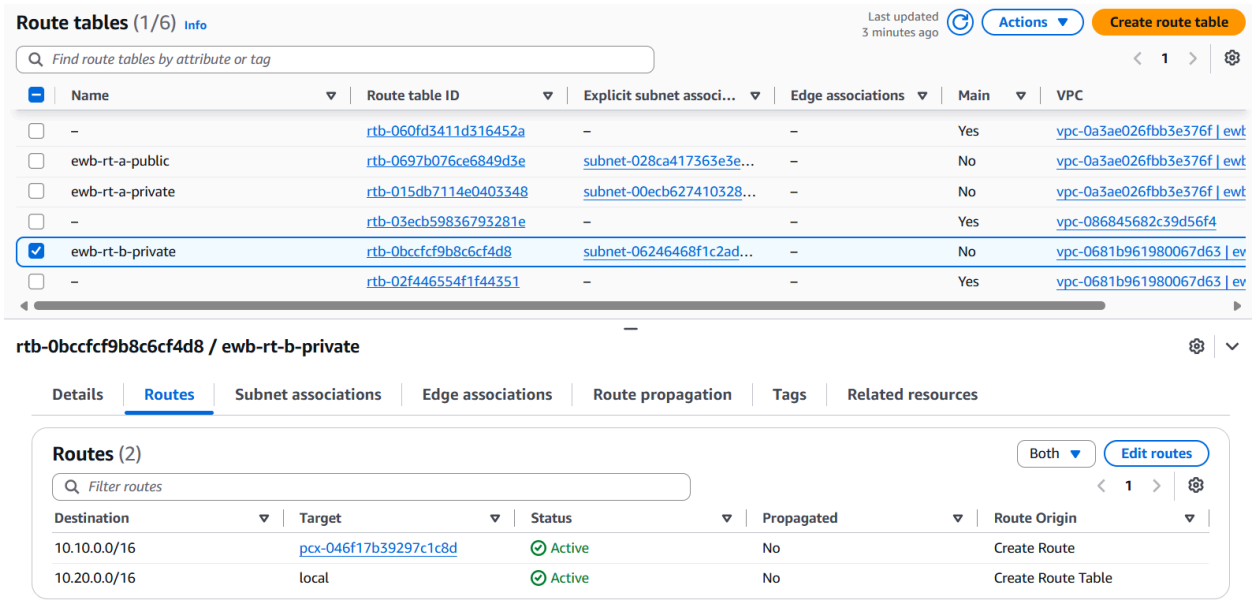
4. **Checkpoint_03_Active_Peering.png**:



Updated Route Tables with Peering Targets:

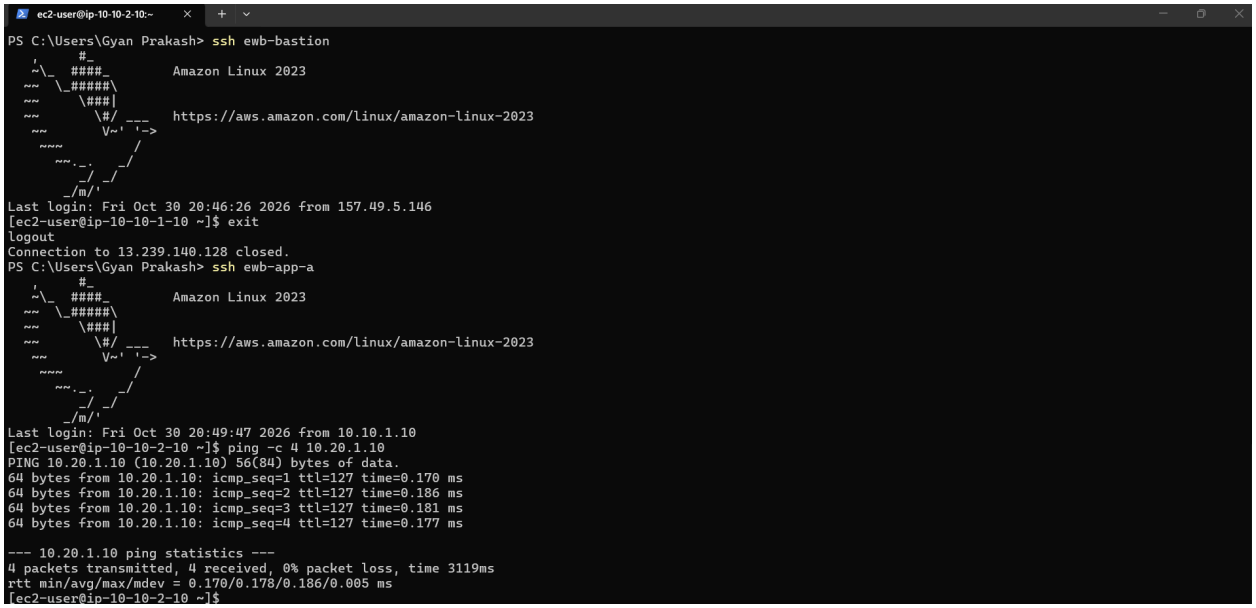

Route tables (1/6) Info
Last updated 1 minute ago
Actions
Create route table
Find route tables by attribute or tag
Name
Route table ID
Explicit subnet associ...
Edge associations
Main
VPC
– rtb-060fd3411d316452a – – Yes vpc-0a3ae026fbb3e376f | ewb
ewb-rt-a-public rtb-0697b076ce6849d3e subnet-028ca417363e3e... – No vpc-0a3ae026fbb3e376f | ewb
ewb-rt-a-private rtb-015db7114e0403348 subnet-00ecb627410328... – No vpc-0a3ae026fbb3e376f | ewb
– rtb-03ecb59836793281e – – Yes vpc-086845682c39d56f4
ewb-rt-b-private rtb-0bccfcf9b8c6cf4d8 subnet-06246468f1c2ad... – No vpc-0681b961980067d63 | ev
– rtb-02f446554f1f44351 – – Yes vpc-0681b961980067d63 | ev
rtb-0697b076ce6849d3e / ewb-rt-a-public
Details
Routes
Subnet associations
Edge associations
Route propagation
Tags
Related resources
Routes (3)
Both
Edit routes
Filter routes
Destination
Target
Status
Propagated
Route Origin
0.0.0.0/0 igw-0b9672e4f26bc9608 Active No Create Route
10.10.0.0/16 local Active No Create Route Table
10.20.0.0/16 pcx-046f17b39297c1c8d Active No Create Route


Route tables (1/6) Info
Last updated 2 minutes ago
Actions
Create route table
Find route tables by attribute or tag
Name
Route table ID
Explicit subnet associ...
Edge associations
Main
VPC
– rtb-060fd3411d316452a – – Yes vpc-0a3ae026fbb3e376f | ewb
ewb-rt-a-public rtb-0697b076ce6849d3e subnet-028ca417363e3e... – No vpc-0a3ae026fbb3e376f | ewb
ewb-rt-a-private rtb-015db7114e0403348 subnet-00ecb627410328... – No vpc-0a3ae026fbb3e376f | ewb
– rtb-03ecb59836793281e – – Yes vpc-086845682c39d56f4
ewb-rt-b-private rtb-0bccfcf9b8c6cf4d8 subnet-06246468f1c2ad... – No vpc-0681b961980067d63 | ev
– rtb-02f446554f1f44351 – – Yes vpc-0681b961980067d63 | ev
rtb-015db7114e0403348 / ewb-rt-a-private
Details
Routes
Subnet associations
Edge associations
Route propagation
Tags
Related resources
Routes (2)
Both
Edit routes
Filter routes
Destination
Target
Status
Propagated
Route Origin
10.10.0.0/16 local Active No Create Route Table
10.20.0.0/16 pcx-046f17b39297c1c8d Active No Create Route

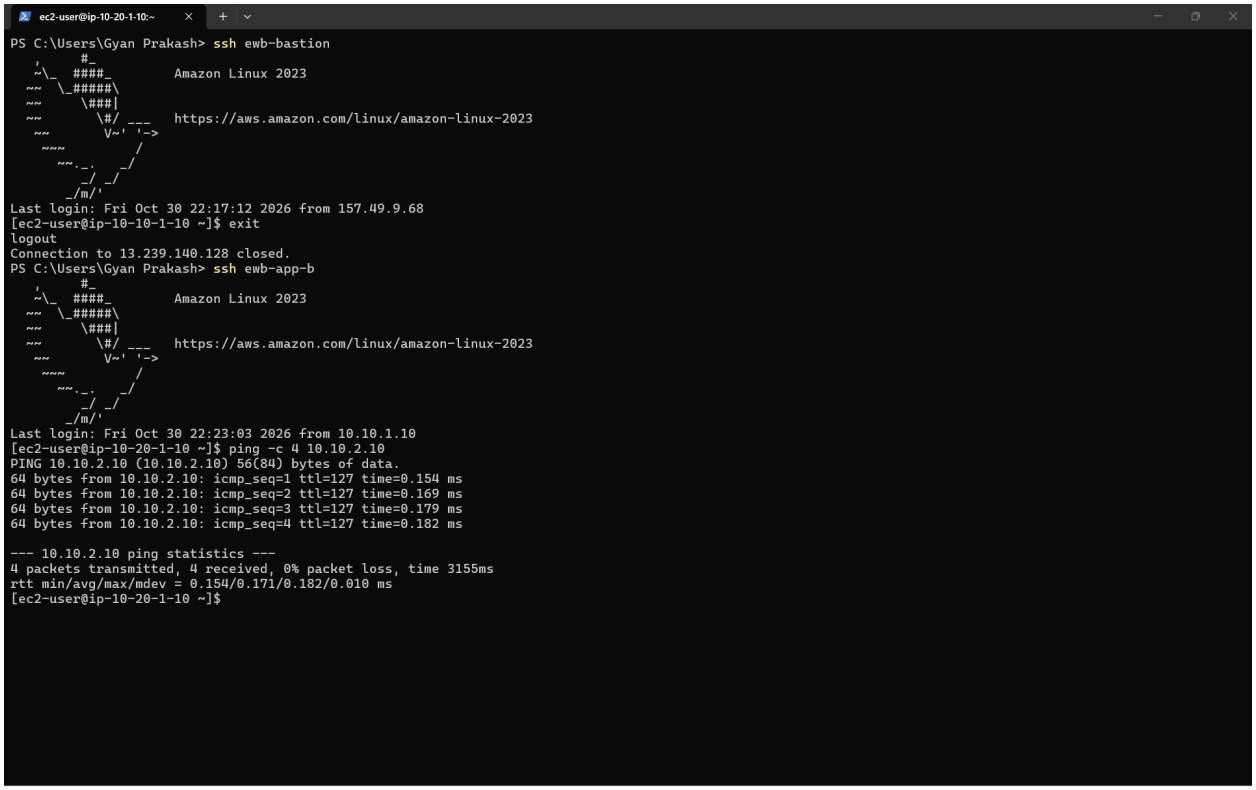


C. Post-Peering Verification Success

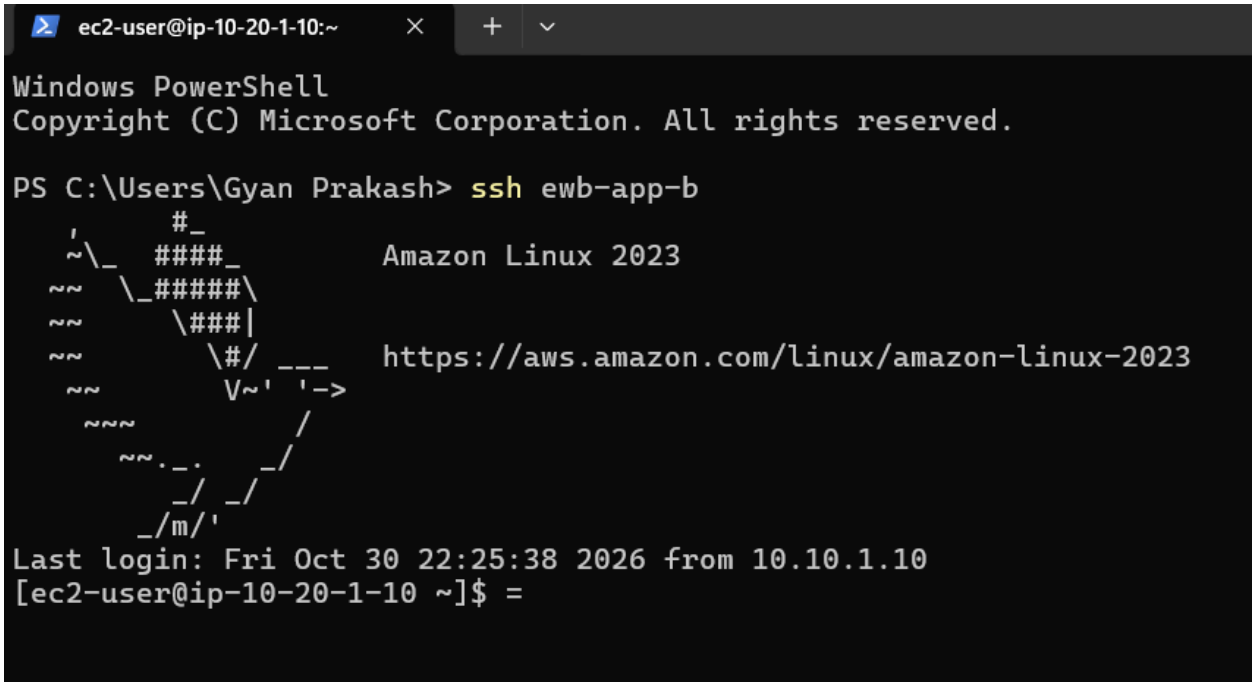
Forward Path Ping Test (ewb-app-a to ewb-app-b):



Reverse Path Ping Test (ewb-app-b to ewb-app-a):



Private SSH Access Test:

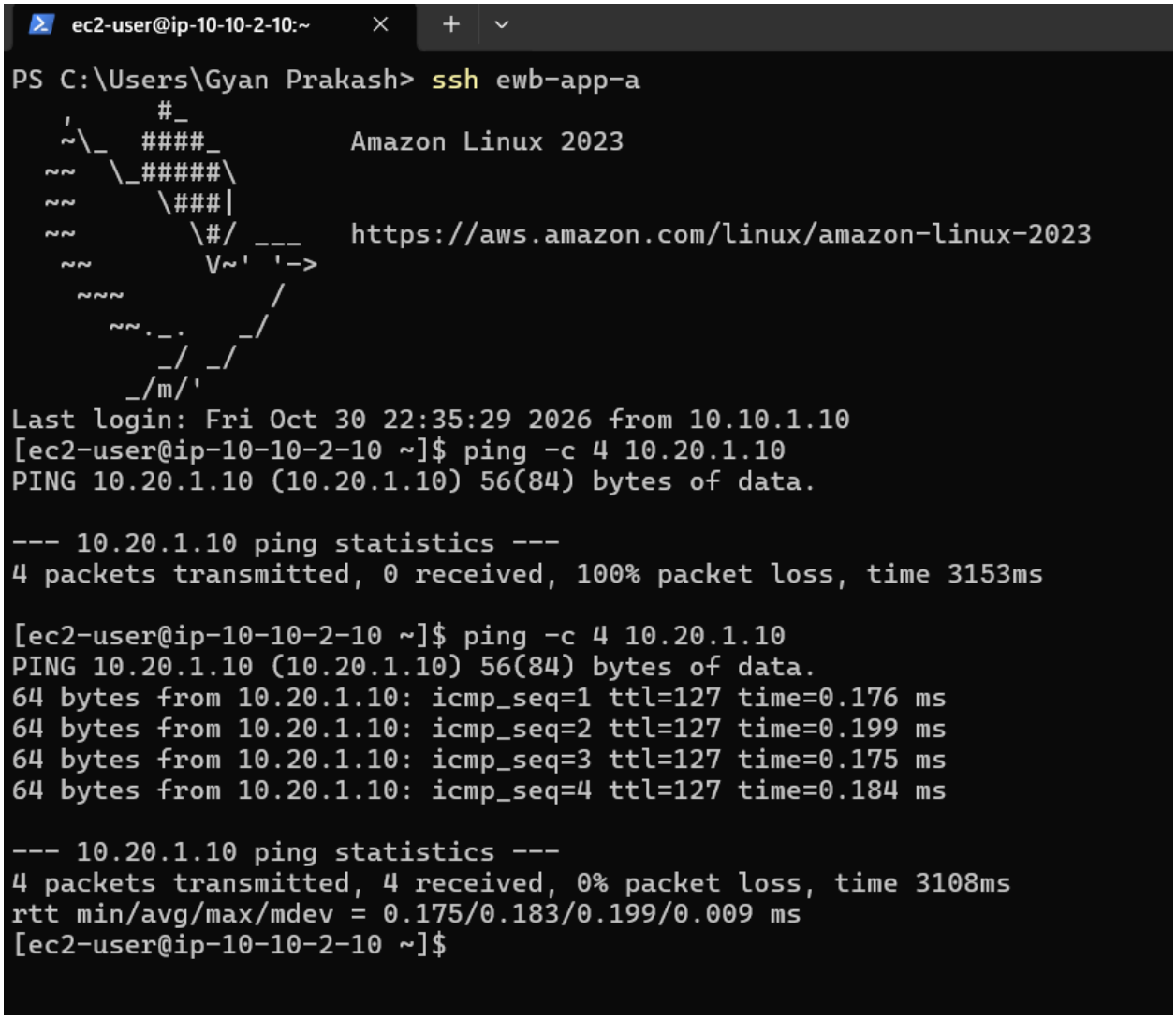


5. Controlled Failure & Routing Explanation

Controlled Failure Scenario:

To demonstrate the operational dependency on route table configuration, the route pointing destination 10.20.0.0/16 to pcx-046f17b39297c1c8d was temporarily removed from ewb-rt-a-private. Ping tests from ewb-app-a to ewb-app-b immediately failed with 100% packet loss. Upon restoring the route target in the route table, ping traffic recovered instantly without requiring any changes to the peering connection or instances.

Controlled Failure Evidence (Test_06_Controlled_Failure_Recovery.png):

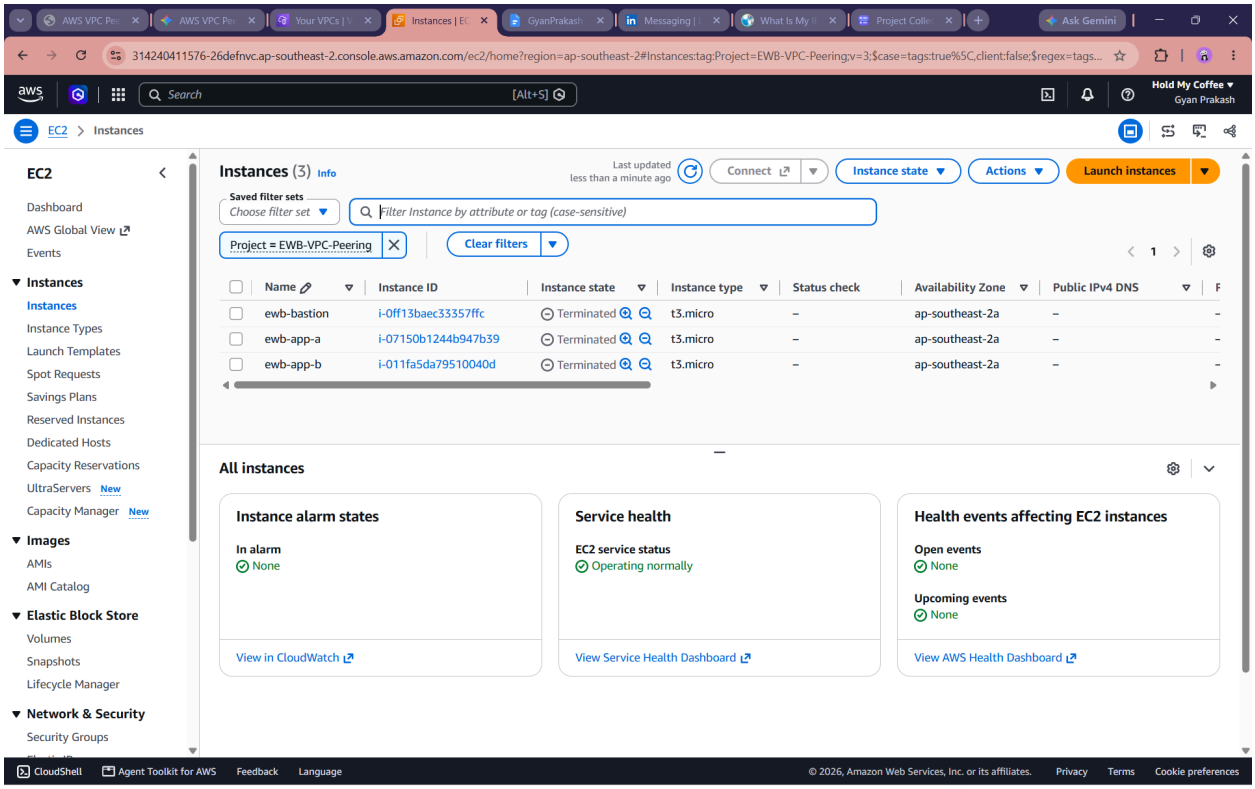


6. Cleanup Evidence & Key Learnings

Resource Teardown Verification:

Following successful verification, all lab resources (EC2 instances ewb-bastion, ewb-app-a, and ewb-app-b) were terminated to prevent unnecessary cloud spend. The VPC peering connection and associated route tables were subsequently deleted.

Cleanup Evidence (Cleanup_Evidence.png):



Key Architectural Learnings:

10. **Bi-directional Routing Requirement:** VPC Peering active status alone is insufficient; both requester and accepter route tables must explicitly route target CIDRs to the peering ID.
11. **Security Group Layers:** Security groups must permit incoming ICMP/SSH traffic from the remote VPC CIDR or specific peer security groups.

Non-Transitive Nature: AWS VPC Peering does not support transitive routing; each VPC pair requires a direct peering relationship.